

Course Code	Course Title	L	T	P	C
BCSE404L	Internet and Web Programming	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To provide a practical approach to learning web technologies and programming. 2. To enable full-stack web development by learning every aspect of building a database driven web application. 3. To demonstrate how the client-server model of Internet programming works					
Course Outcomes					
After successfully completing the course the student should be able to					
1. Identify the different protocols used in the web and comprehend the architecture of the web. 2. Apply frontend technologies such as markup, styling and interactivity to the web pages. 3. Build full database-driven web applications using server-side programming. 4. Create asynchronous processes thereby creating effective web pages. 5. Construct complete websites using the latest web technologies and MVC pattern					
Module:1	Introduction To Internet	3 hours			
Internet Overview- Networks – WWW –Web Protocols – Web Organization and Addressing – Internet Service Providers, DNS Servers, Connection Types, Internet Addresses - Web Browsers and Web Servers -Security and Vulnerability-Web System Architecture – URL - Domain Name – Web Content Authoring - Webserver Administration – Search Engines					
Module:2	Web Designing	8 hours			
HTML5 – Text tags; Graphics, Video and Sound Tags; Link and Anchor Tags; Table Tags; Form elements, HTML 5 Input types, semantic tags, CSS3 - Selectors, Box Model, Backgrounds and Borders, Text Effects, Animations, Multiple Column Layouts					
Module:3	Client Side Scripting	8 hours			
JavaScript -Variables and Data Types - Statements – Operators- Literals- Functions- Objects- Arrays- Built-in Objects - Exceptions, Event handling, Validation - DOM - Canvas- JQuery					
Module:4	Server Side Scripting	6 hours			
Structure of PHP – Expressions and Control Flow – Functions and Objects – Arrays – Form Handling – File Handling – Email - Validation and Error Handling – Cookies – Session					
Module:5	Database Connectivity	6 hours			
MySQL – Introduction - database design concepts - the Structured Query, Language (SQL) - communicating with a MySQL backend via the PHP - MySQL API - More MySQL database access – PHP Data Objects - JSON					
Module:6	Developing Interactive Web Applications	6 hours			
XML Basics – Namespaces – Transforming XML documents - XSL, XSLT, XML Schema- DTD,XSD - AJAX –AJAX calls - XML http – request – response – AJAX					

with PHP - Data Formats - AJAX with Database – Processing Server Response - AJAX Security			
Module:7	Application Development		6 hours
Introduction to Node.js- NPM - Events, Timers, and Callbacks in Node.js – file upload – email – Express framework – request –response –routing - templates- view engines. Introduction to Mongo DB- creating DB, collection – CRUD operations - Accessing MongoDB from Node.js. – Accessing online Mongo DB from Node JS.			
Module:8	Recent Trends		2 hours
Guest lectures from Industry and, Research and Development Organizations			
	Total Lecture hours:		45 hours
Text Book(s)			
1.	Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, "Internet & World Wide Web How to Program", Deitel series, 5th edition, 2012		
2.	Simon Holmes, GETTING MEAN: with MONGO,EXPRESS, ANGLUAR JS, & NODE JS, Second edition, Manning Publications, 2015		
Reference Books			
1.	DT Editorial Services, HTML 5 Black Book, 2nd edition, Dream Tech press, 2018.		
Mode of Evaluation: CAT, Assignments, Quiz and FAT			
Recommended by Board of Studies		12-05-2023	
Approved by Academic Council		No. 70	Date 24-06-2023